

Graph in $\mathbb{R}^3$ and Level Curves

Definition (Graph in $\mathbb{R}^3$)

The **graph** of function $z = f(x, y)$ with domain D is:

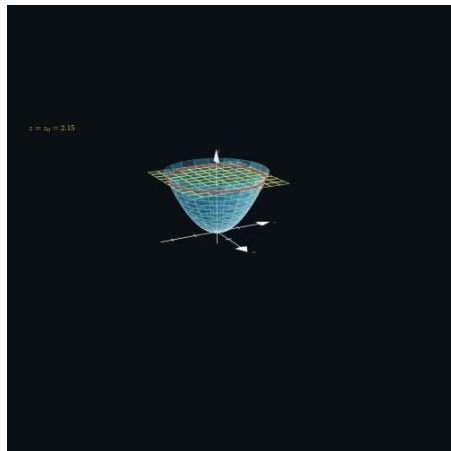
$$\{(x, y, z) \in \mathbb{R}^3 \mid (x, y) \in D \text{ and } z = f(x, y)\}.$$

It is a **surface** defined by the function.

Question: How can we plot a surface?

- For $s = g(t)$, we connect sample points with a smooth curve.
- For $z = f(x, y)$, this fails. Instead, for each $z = z_0$, draw the **level curve** $z_0 = f(x, y)$.
- This is the **intersection** of the surface and horizontal plane $z = z_0$:

$$L_{z_0} := \{(x, y) \mid f(x, y) = z_0\}.$$



Slicing $z = f(x, y)$ with horizontal plane $z = z_0$ to obtain level curves